

Lecture 3 Notes - Characteristics of Time Series

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- **Reading:** Chapter 1.3-1.7 – Shumway and Stoffer

1 Review / basic concepts

- Stochastic process
- White noise
- Moving average
- Autoregression
- Random walk (w/ and w/o drift)
- Signals in noise

2 This week - Measures of dependence

- Mean function
- Autocovariance
- Autocorrelation
- Cross-covariance
- Cross-correlation
- Stationarity
- Multidimensional time series
- Random number generation

3 Mean and variance

Last week and in your lab, we saw the concept of how mean and variance differ for a white noise process, a random walk, and a random walk with drift. The *mean* and *variance functions* of a time series are useful descriptors because they helps us determine something about the drift and the spread of the data that we should expect over time.

The *mean function* is defined as $\mu_{xt} = \mathbb{E}(x_t)$

The *variance function* is $\sigma_t^2 = \text{Var}(x_t) = \mathbb{E}[(x_t - \mu_t)^2]$.

Let's revisit some examples from before:

3.1 White noise

For a white noise time series, $\mu_{wt} = \mathbb{E}(w_t) = 0$ for all t . $\text{Var}(w_t) = 1$ (for a Gaussian white noise series).

3.2 Moving average

What if we apply a 3-point moving average? Although this induces some correlation structure, it actually does not change the mean function at all:

$$\mu_{vt} = \mathbb{E}(v_t) = \frac{1}{3}[\mathbb{E}(w_{t-1}) + \mathbb{E}(w_t) + \mathbb{E}(w_{t+1})] = 0$$

3.3 Random walk with drift

For the random walk with drift, the mean function is just the line $\mu_{xt} = \delta t + \sum_{j=1}^t \mathbb{E}(w_j) = \delta t$.

3.4 Signal plus noise

And for a signal plus noise (as an example, we'll use this sinusoid):

$$\begin{aligned}\mu_{xt} = \mathbb{E}(x_t) &= \mathbb{E}[A \cos(2\pi\omega t + \phi) + w_t] \\ &= \mathbb{E}[A \cos(2\pi\omega t + \phi)] + \mathbb{E}[w_t] \\ &= \mathbb{E}[A \cos(2\pi\omega t + \phi)]\end{aligned}\tag{1}$$

4 Autocovariance

What if we instead want to know something about the dependence between two points s and t within the same time series. We'll call this *autocovariance*, γ .

$$\gamma_x(s, t) = \text{cov}(x_s, x_t) = \mathbb{E}[(x_s - \mu_s)(x_t - \mu_t)]$$

When $s = t$, we have:

$$\gamma_x(t, t) = \mathbb{E}[(x_t - \mu_t)^2] = \text{Var}(x_t)$$

For white noise, there should be no dependence between differing time points. By definition, $\mathbb{E}(w_t) = 0$ and:

$$\gamma_w(s, t) = \text{cov}(w_s, w_t) = \begin{cases} \sigma_w^2, & \text{if } s = t, \\ 0, & s \neq t \end{cases} \quad (2)$$

4.1 Covariance of linear combinations

If we have random variables $U = \sum_{j=1}^m a_j X_j$ and $V = \sum_{k=1}^r b_k Y_k$ that are linear combinations of (finite variance) random variables X_j and Y_k , then the covariance of these is:

$$\text{cov}(U, V) = \sum_{j=1}^m \sum_{k=1}^r a_j b_k \text{cov}(X_j, Y_k)$$

Also, $\text{var}(U) = \text{cov}(U, U)$.

So how can we use this? Now we can try this for the moving average example.

$$\begin{aligned} \gamma_v(s, t) = \text{cov}(v_s, v_t) &= \text{cov}\left(\frac{1}{3}(w_{s-1} + w_s + w_{s+1}), \frac{1}{3}(w_{t-1} + w_t + w_{t+1})\right) \\ &= \frac{1}{9} \text{cov}(w_{s-1} + w_s + w_{s+1}, w_{t-1} + w_t + w_{t+1}) \end{aligned} \quad (3)$$

When $s = t$, we have:

$$\begin{aligned} \gamma_v(t, t) &= \text{cov}(v_t, v_t) \\ &= \text{cov}\left(\frac{1}{3}(w_{t-1} + w_t + w_{t+1}), \frac{1}{3}(w_{t-1} + w_t + w_{t+1})\right) \\ &= \frac{1}{9}(\text{cov}(w_{t-1}, w_{t-1}) + \text{cov}(w_t, w_t) + \text{cov}(w_{t+1}, w_{t+1}) \\ &\quad + 2 \text{cov}(w_{t-1}, w_t) + 2 \text{cov}(w_t, w_{t+1}) + 2 \text{cov}(w_{t-1}, w_{t+1})) \\ &= \frac{1}{9}(\sigma_w^2 + \sigma_w^2 + \sigma_w^2 + 0 + 0 + 0) \\ &= \frac{3}{9} \sigma_w^2 \\ &= \frac{1}{3} \sigma_w^2 \end{aligned} \quad (4)$$

When $s = t + 1$, we have:

$$\begin{aligned} \gamma_v(t+1, t) &= \text{cov}\left(\frac{1}{3}(w_t + w_{t+1} + w_{t+2}), \frac{1}{3}(w_{t-1} + w_t + w_{t+1})\right) \\ &= \frac{1}{9}(\text{cov}(w_t, w_t) \\ &\quad + \text{cov}(w_{t+1}, w_{t+1}) \\ &\quad + \text{cov}(w_t, w_{t-1}) \\ &\quad + \text{cov}(w_t, w_{t+1}) \\ &\quad + \text{cov}(w_{t+1}, w_{t-1}) \\ &\quad + \text{cov}(w_{t+1}, w_t) \\ &\quad + \text{cov}(w_{t+2}, w_{t-1}) \\ &\quad + \text{cov}(w_{t+2}, w_t) \\ &\quad + \text{cov}(w_{t+2}, w_{t+1})) \\ &= \frac{1}{9}(\text{cov}(w_t, w_t) + \text{cov}(w_{t+1}, w_{t+1})) \\ &= \frac{2}{9} \sigma_w^2 \end{aligned} \quad (5)$$

If we then follow this for more lags, we get:

$$\gamma_v(s, t) = \begin{cases} \frac{3}{9}\sigma_w^2, & \text{if } s = t, \\ \frac{2}{9}\sigma_w^2, & \text{if } |s - t| = 1, \\ \frac{1}{9}\sigma_w^2, & \text{if } |s - t| = 2, \\ 0, & \text{if } |s - t| > 2 \end{cases} \quad (6)$$

Why is this interesting? The autocovariance depends only on the *lag* between s and t and not on the absolute location of these time points. We'll come back to this when we talk about *stationarity*.